

MA 104: TUTORIAL-6(A) SOLUTIONS

Example 0.1. Find a basis of solutions of the ODE

$$(1 - x^2)y'' - xy' + p^2y = 0.$$

Show that if p is a positive integer, then there exists a polynomial of degree p which satisfies the given ODE.

Solution. Given

$$(0.1) \quad (1 - x^2)y'' - xy' + p^2y = 0$$

Take

$$(0.2) \quad y = \sum_{m=0}^{\infty} a_m x^m$$

Then

$$(0.3) \quad y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$$

and

$$(0.4) \quad y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$$

Substitute equations (0.2), (0.3), (0.4) in (0.1), we get

$$\sum_{m=2}^{\infty} m(m-1) a_m x^{m-2} - \sum_{m=2}^{\infty} m(m-1) a_m x^m - \sum_{m=1}^{\infty} m a_m x^m + \sum_{m=0}^{\infty} p^2 a_m x^m = 0$$

Substituting $s = m - 2$ in the first series and $s = m$ in the above series, we get

$$(0.5) \quad \sum_{s=0}^{\infty} (s+2)(s+1) a_{s+2} x^s - \sum_{s=2}^{\infty} s(s-1) a_s x^s - \sum_{s=1}^{\infty} s a_s x^s + \sum_{s=0}^{\infty} p^2 a_s x^s = 0$$

Comparing co-efficients of x^0 in (0.5), we get

$$2a_2 + p^2 a_0 = 0 \quad \Rightarrow \quad a_2 = -\frac{p^2}{2} a_0$$

Comparing co-efficients of x^1 in (0.5), we get

$$6a_3 - a_1 + p^2 a_1 = 0 \quad \Rightarrow \quad a_3 = \frac{(1 - p^2)}{6} a_1$$

Comparing co-efficients of x^s in (0.5), $s \geq 1$, we get

$$(s+2)(s+1) a_{s+2} - s(s-1) a_s - s a_s + p^2 a_s = 0$$

implies

$$(0.6) \quad a_{s+2} = \frac{s^2 - p^2}{(s+2)(s+1)} a_s$$

Letting $s = 0, 2, 4, \dots, 2m - 2$ in (0.6), we get

$$\begin{aligned} a_2 &= \frac{-p^2}{2} a_0, \\ a_4 &= \frac{4-p^2}{4 \cdot 3} a_2 = \frac{(4-p^2)(0-p^2)}{4!} a_0, \\ a_6 &= \frac{4^2-p^2}{6 \cdot 5} a_4 = \frac{(4^2-p^2)(2^2-p^2)(0-p^2)}{6!} a_0, \\ &\vdots \\ a_{2m} &= \frac{((2m-2)^2-p^2) \cdots (0-p^2)}{2m!} a_0 = \frac{\prod_{j=1}^m ((2m-2j)^2-p^2)}{2m!} a_0 \end{aligned}$$

In the same way, letting $s = 1, 3, 5, \dots, 2m - 1$ in (0.6), we get

$$\begin{aligned} a_3 &= \frac{(1-p^2)}{3!} a_1, \\ a_5 &= \frac{3^2-p^2}{5 \cdot 4} a_3 = \frac{(3^2-p^2)(1-p^2)}{5!} a_1, \\ &\vdots \\ a_{2m+1} &= \frac{\prod_{j=1}^m [(2m+1-2j)^2-p^2]}{(2m+1)!} a_1 \end{aligned}$$

Therefore,

$$\begin{aligned} y(x) &= a_0 + a_1 x + a_2 x^2 + a_3 x^3 + a_4 x^4 + a_5 x^5 + a_6 x^6 + \cdots \\ &= [a_0 + a_2 x^2 + a_4 x^4 + a_6 x^6 + \cdots] + [a_1 x + a_3 x^3 + a_5 x^5 + \cdots] \\ &= a_0 \left[1 + \frac{(0-p^2)}{2!} x^2 + \frac{(2^2-p^2)(0-p^2)}{4!} x^4 + \cdots \right] + a_1 \left[x + \frac{1-p^2}{6} x^3 + \frac{(1-p^2)(3^2-p^2)}{5!} x^5 + \cdots \right] \\ &:= a_0 y_1(x) + a_1 y_2(x) \end{aligned}$$

Hence

$$\begin{aligned} y_1(x) &= 1 + \frac{(0-p^2)}{2!} x^2 + \frac{(0-p^2)(2^2-p^2)}{4!} x^4 + \cdots \\ y_2(x) &= x + \frac{(1-p^2)}{3!} x^3 + \frac{(1-p^2)(3^2-p^2)}{5!} x^5 + \cdots \end{aligned}$$

form a basis of solutions.

Note that if p is an even integer then $y_1(x)$ is a polynomial of degree p and if p is an odd integer then $y_2(x)$ is a polynomial of degree p . (because coefficient of $x^s, s > p$ is $\frac{(0-p^2)(2^2-p^2) \cdots (p^2-p^2) \cdots ((s-2)^2-p^2)}{s!}$ OR $\frac{(1-p^2)(3^2-p^2) \cdots (p^2-p^2) \cdots ((s-2)^2-p^2)}{s!}$ respectively as s is even or odd.)

Example 0.2. Express $\sin^{-1} x$ in the form of a power series

$$\sum_{n=0}^{\infty} a_n x^n$$

by solving

$$y' = (1 - x^2)^{-1/2}.$$

Use this result to obtain the formula

$$\frac{\pi}{6} = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3 \cdot 2^3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{5 \cdot 2^5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \cdot \frac{1}{7 \cdot 2^7} + \dots$$

Solution. We will solve

$$(0.7) \quad y' = (1 - x^2)^{-1/2}$$

by power series method.

Take

$$(0.8) \quad y = \sum_{m=0}^{\infty} a_m x^m$$

Then

$$(0.9) \quad y' = \sum_{m=1}^{\infty} m a_m x^{m-1} = \sum_{s=0}^{\infty} (s+1) a_{s+1} x^s$$

Next,

$$(0.10) \quad \begin{aligned} (1 - x^2)^{-1/2} &= 1 - \frac{(-x^2)}{2} + \frac{1 \cdot 3}{2^2 \cdot 2!} (-x^2)^2 - \frac{1 \cdot 3 \cdot 5}{2^3 \cdot 3!} (-x^2)^3 + \dots + (-1)^m \frac{1 \cdot 3 \cdot 5 \dots (2m-1)}{2^m \cdot m!} (-x^2)^m + \dots \\ &= 1 + \frac{x^2}{2} + \frac{1 \cdot 3}{2^2 \cdot 2!} x^4 + \frac{1 \cdot 3 \cdot 5}{2^3 \cdot 3!} x^6 + \dots + \frac{1 \cdot 3 \cdot 5 \dots (2m-1)}{2^m \cdot m!} x^{2m} + \dots \end{aligned}$$

Substitute equations (0.9), (0.10) in (0.7), we get

$$\begin{aligned} y' &= (1 - x^2)^{-1/2} \\ \sum_{s=0}^{\infty} (s+1) a_{s+1} x^s &= \sum_{m=0}^{\infty} \frac{1 \cdot 3 \dots (2m-1)}{2^m \cdot m!} x^{2m} \end{aligned}$$

Comparing co-efficients of x^{2m-1} and x^{2m} in above equation, we get

$$\begin{aligned} a_{2m} &= 0, \quad \forall m \geq 0 \\ a_{2m+1} &= \frac{1 \cdot 3 \dots (2m-1)}{(2m+1) \cdot 2^m \cdot m!} = \frac{1 \cdot 3 \dots (2m-1)}{2 \cdot 4 \dots 2m} \cdot \frac{1}{2m+1}, \quad \forall m \geq 0 \end{aligned}$$

Therefore from (0.8), The solution is

$$y(x) = \sum_{m=0}^{\infty} \frac{1 \cdot 3 \dots (2m-1)}{2 \cdot 4 \dots 2m} \cdot \frac{1}{2m+1} x^{2m+1}$$

Since $\sin^{-1} x$ is also a solution of (0.7), we have

$$\sin^{-1} x = y(x) + C = C + \sum_{m=0}^{\infty} \frac{1 \cdot 3 \dots (2m-1)}{2 \cdot 4 \dots 2m} \cdot \frac{1}{2m+1} x^{2m+1}$$

By taking $x = 0$, we get

$$C = 0.$$

$$\sin^{-1} x = \sum_{m=0}^{\infty} \frac{1 \cdot 3 \dots (2m-1)}{2 \cdot 4 \dots (2m)} \cdot \frac{1}{2m+1} x^{2m+1}.$$

Hence putting $x = \frac{1}{2}$, we get

$$\begin{aligned} \frac{\pi}{6} &= \sum_{m=0}^{\infty} \frac{1 \cdot 3 \dots (2m-1)}{2 \cdot 4 \dots (2m)} \cdot \frac{1}{2m+1} \frac{1}{2^{2m+1}} \\ &= \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3 \cdot 2^3} + \frac{1 \cdot 3}{2 \cdot 4} \cdot \frac{1}{5 \cdot 2^5} + \dots \end{aligned}$$

Example 0.3. Show that for $m < n$ as well as $m - n = 2k + 1$, where k is a non-negative integer,

$$\int_{-1}^1 x^m P_n(x) dx = 0.$$

Solition. Note that for any $k < n$, we know that

$$(0.11) \quad \frac{d^k}{dx^k} (x^2 - 1)^n|_{x=\pm 1} = 0.$$

(0.11) can be verify using following three steps:

- $(x^2 - 1)^n = (x - 1)^n (x + 1)^n$

- Leibniz Rule:

$$\frac{d^k}{dx^k} [u(x)v(x)] = \sum_{j=0}^k \binom{k}{j} u^{(k-j)}(x) v^{(j)}(x)$$

- The k-the derivative of a term $(x - a)^n$ is given by:

$$\frac{d^k}{dx^k} (x - a)^n = \frac{n!}{(n - k)!} (x - a)^{n-k}$$

and this derivative is equal to 0 at $x = a$ if $k < n$.

Using Rodrigus formula and integration by parts, we have

$$\begin{aligned} I &= \int_{-1}^1 x^m P_n(x) dx = \int_{-1}^1 x^m \frac{1}{2^n n!} \left(\frac{d^n}{dx^n} (x^2 - 1)^n \right) dx \\ &= \frac{1}{2^n n!} \left[x^m \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n \right]_{-1}^1 - \frac{m}{2^n n!} \int_{-1}^1 x^{m-1} \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n dx \\ &= \frac{-m}{2^n n!} \int_{-1}^1 x^{m-1} \frac{d^{n-1}}{dx^{n-1}} (x^2 - 1)^n dx \end{aligned}$$

Repeating this m -times, we have

$$\begin{aligned} I &= (-1)^m \frac{m(m-1) \dots 1}{2^n n!} \int_{-1}^1 \frac{d^{n-m}}{dx^{n-m}} (x^2 - 1)^n dx \\ &= (-1)^m \frac{m!}{2^n \cdot n!} \left[\frac{d^{n-m-1}}{dx^{n-m-1}} (x^2 - 1)^n \right]_{-1}^1 \\ &= 0. \end{aligned}$$

If $m - n = 2k + 1$ then

$x^m P_n$ is an odd function.

(because $(-x)^m P_n(-x) = (-1)^m (-1)^n x^m P_n(x) = (-1)^{m+n} x^m P_n(x) = -x^m P_n(x)$, $m+n = (m-n)+2n : \text{odd} + \text{even}$)

Therefore,

$$\int_{-1}^1 x^m P_n(x) dx = 0.$$

Example 0.4. The generating function of a sequence $\{f_n\}_{n \geq 0}$ is the function

$$G(x, t) = \sum_{n=0}^{\infty} f_n t^n.$$

Show

$$G(x, t) := \frac{1}{\sqrt{1 - 2xt + t^2}} = \sum_{n=0}^{\infty} P_n(x) t^n$$

is the generating function of the Legendre polynomials $\{P_n(x)\}_{n \geq 0}$. Using this prove that

- (1) $(2n + 1)xP_n(x) = (n + 1)P_{n+1}(x) + nP_{n-1}(x)$,
- (2) $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$.

Solution. Given that

$$\begin{aligned} G(x, t) &:= (1 - 2xt + t^2)^{-1/2} \\ &= (1 - t(2x - t))^{-1/2} \\ &= \sum_{n=0}^{\infty} \binom{-1/2}{n} (-t(2x - t))^n \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n (2n)!}{2^{2n} (n!)^2} (-1)^n t^n (2x - t)^n \\ &= \sum_{n=0}^{\infty} \frac{(2n)!}{2^{2n} (n!)^2} t^n \sum_{k=0}^n \binom{n}{k} (2x)^{n-k} (-t)^k \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{(2n)!}{2^{2n} (n!)^2} \frac{n!}{(n-k)!k!} (-1)^k (2x)^{n-k} t^{n+k} \\ &\quad \text{substitute } m=n+k \text{ (limit for second summation: } k \leq n = m - k \Rightarrow 2k \leq m) \\ &= \sum_{m=0}^{\infty} \left[\sum_{k=0}^{\lfloor m/2 \rfloor} \frac{(2m-2k)!}{2^{2(m-k)} ((m-k)!)^2} \frac{(m-k)!}{(m-2k)!k!} (-1)^k (2x)^{m-2k} \right] t^m \\ &= \sum_{m=0}^{\infty} \left[\sum_{k=0}^{\lfloor m/2 \rfloor} (-1)^k \frac{(2m-2k)!}{2^m k!} \frac{1}{(m-2k)!(m-k)!} x^{m-2k} \right] t^m \\ &= \sum_{m=0}^{\infty} P_m(x) t^m \end{aligned}$$

Solution of part (1). We have

$$(0.12) \quad (1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x) t^{n-1}$$

Differentiating w.r.t. t , we get

$$-\frac{1}{2}(1 - 2xt + t^2)^{-3/2}(-2x + 2t) = \sum_{n=0}^{\infty} nP_n(x) t^{n-1}$$

Multiplying by $(1 - 2xt + t^2)$ on both sides, we have

$$\begin{aligned} (x - t)(1 - 2xt + t^2)^{-1/2} &= (1 - 2xt + t^2) \sum_{n=0}^{\infty} nP_n(x) t^{n-1} \\ (x - t) \sum_{n=0}^{\infty} P_n(x) t^n &= (1 - 2xt + t^2) \sum_{n=0}^{\infty} nP_n(x) t^{n-1} \quad (\text{Using (0.12)}) \end{aligned}$$

The coefficient of t^n on L.H.S. $= xP_n(x) - P_{n-1}(x)$

The coefficient of t^n on R.H.S. $= (n + 1)P_{n+1}(x) - 2nP_n(x) + (n - 1)P_{n-1}(x)$

comparing both this coefficients, we obtain

$$\begin{aligned} xP_n(x) - P_{n-1}(x) &= (n+1)P_{n+1}(x) - 2nxP_n(x) + (n-1)P_{n-1}(x) \\ (2n+1)xP_n(x) &= (n+1)P_{n+1}(x) + nP_{n-1}(x) \end{aligned}$$

Solution of part (2). We have

$$(1 - 2xt + t^2)^{-1/2} = \sum_{n=0}^{\infty} P_n(x)t^n$$

Differentiating w.r.t. t , we get

$$(0.13) \quad (1 - 2xt + t^2)^{-3/2}(x - t) = \sum_{n=0}^{\infty} nP_n(x)t^{n-1}$$

Differentiating w.r.t. x , we get

$$t(1 - 2xt + t^2)^{-3/2} = \sum_{n=0}^{\infty} P'_n(x)t^n$$

Multiplying by $(x - t)$ on both sides, we have

$$t(x - t)(1 - 2xt + t^2)^{-3/2} = (x - t) \sum_{n=0}^{\infty} P'_n(x)t^n$$

$$t \sum_{n=0}^{\infty} nP_n(x)t^{n-1} = (x - t) \sum_{n=0}^{\infty} P'_n(x)t^n \quad (\text{Using (0.13)})$$

$$\sum_{n=0}^{\infty} nP_n(x)t^n = \sum_{n=0}^{\infty} xP'_n(x)t^n - \sum_{n=0}^{\infty} P'_n(x)t^{n+1}$$

Equating the coefficients of t^n , We get

$$nP_n(x) = xP'_n(x) - P'_{n-1}(x)$$

Example 0.5. Show that for $n \neq m$,

$$\int_{-1}^1 (1 - x^2)P'_m(x)P'_n(x) dx = 0.$$

Solution. Integrating by parts, we have

$$\begin{aligned} & \int_{-1}^1 (1 - x^2)P'_m P'_n dx \\ &= [(1 - x^2)P'_m P_n]_{-1}^1 - \int_{-1}^1 P_n \frac{d}{dx} [(1 - x^2)P'_m] dx \\ &= 0 - \int_{-1}^1 P_n [(1 - x^2)P''_m - 2xP'_m] dx \\ &= - \int_{-1}^1 P_n [(m(m+1))P_m] dx \quad (\text{since } P_m \text{ satisfies the Legendre ODE}) \\ &= -m(m+1) \int_{-1}^1 P_m P_n dx \\ &= 0 \quad (\text{by orthogonality relation}). \end{aligned}$$

Example 0.6. Show that under the transformation

$$y(x) = u(x) \exp \left(-\frac{1}{2} \int p(x) dx \right),$$

the equation

$$y'' + p(x)y' + q(x)y = 0$$

reduces to

$$u'' + Q(x)u = 0,$$

where

$$Q(x) = q(x) - \frac{p^2(x)}{4} - \frac{p'(x)}{2}.$$

Use the above method to solve

$$x^2 y'' + xy' + \left(x^2 - \frac{1}{4} \right) y = 0, \quad x > 0.$$

Solution. Given

$$y(x) = u(x) \exp \left(-\frac{1}{2} \int p(x) dx \right)$$

implies

$$u(x) = y(x) \exp \left(\frac{1}{2} \int p(x) dx \right)$$

Differentiating u in above equation w.r.t. x , we get

$$\begin{aligned} u'(x) &= y'(x) \exp \left(\frac{1}{2} \int p(x) dx \right) + y(x) \exp \left(\frac{1}{2} \int p(x) dx \right) \left(\frac{1}{2} p(x) \right) \\ &= \frac{1}{2} \exp \left(\frac{1}{2} \int p(x) dx \right) [2y'(x) + y(x)p(x)] \end{aligned}$$

Again, differentiating u' in above equation w.r.t. x , we get

$$\begin{aligned} u''(x) &= \exp \left(\frac{1}{2} \int p(x) dx \right) \left(\frac{1}{4} p(x) \right) (y(x)p(x) + 2y'(x)) + \frac{[y'(x)p(x) + y(x)p'(x) + 2y''(x)] \exp \left[\frac{1}{2} \int p(x) dx \right]}{2} \\ &= \exp \left(\frac{1}{2} \int p(x) dx \right) \left(\frac{y(x)p^2(x)}{4} + \frac{y'(x)p(x)}{2} + \frac{y'(x)p(x)}{2} + \frac{y(x)p'(x)}{2} + y''(x) \right) \\ &= \exp \left(\frac{1}{2} \int p(x) dx \right) \left[y''(x) + y'(x)p(x) + y(x) \frac{p'(x)}{2} + y(x) \frac{p^2(x)}{4} \right] \\ &= \exp \left(\frac{1}{2} \int p(x) dx \right) \left[-q(x)y(x) + y(x) \left(\frac{p'(x)}{2} + \frac{p^2(x)}{4} \right) \right] \\ &= y(x) \exp \left(\frac{1}{2} \int p(x) dx \right) \left(\frac{p^2(x)}{4} + \frac{p'(x)}{2} - q(x) \right) \\ &= -Q(x)u. \end{aligned}$$

Hence, we can convert

$$y'' + p(x)y' + q(x)y = 0$$

into

$$u''(x) + Q(x)u = 0.$$

using transformation

$$y(x) = u(x) \exp \left(-\frac{1}{2} \int p(x) dx \right).$$

Where,

$$Q(x) = q(x) - \frac{p^2(x)}{4} - \frac{p'(x)}{2}.$$

Next, Consider

$$x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0$$

$$(0.14) \quad y'' + \frac{y'}{x} + \frac{4x^2 - 1}{4x^2}y = 0$$

Comparing equation (0.14) with $y'' + p(x)y' + q(x)y = 0$

Here

$$q(x) = \frac{4x^2 - 1}{4x^2} \text{ and } p(x) = \frac{1}{x}$$

Hence

$$Q(x) = \frac{4x^2 - 1}{4x^2} - \frac{1}{4x^2} + \frac{1}{2x^2} = 1$$

Therefore by previous discussion we can convert equation (0.14) into the following ODE

$$(0.15) \quad u'' + u = 0$$

Equation (0.15) is a second order ode with constant coefficient and general solution of (0.15) is given by

$$u = A \cos x + B \sin x, \text{ Where A and B are arbitrary constants.}$$

$$y(x) \exp\left(\frac{1}{2} \int p(x) dx\right) = A \cos x + B \sin x.$$

$$\begin{aligned} y(x) &= \exp\left(-\frac{1}{2} \ln x\right)(A \cos x + B \sin x) \\ &= \frac{1}{\sqrt{x}}(A \cos x + B \sin x). \end{aligned}$$

Hence, $\frac{1}{\sqrt{x}} \cos x$ and $\frac{1}{\sqrt{x}} \sin x$ form a basis of the solutions of

$$x^2 y'' + xy' + (x^2 - \frac{1}{4})y = 0.$$

Example 0.7. Find a basis of solutions by the Frobenius method:

- (1) $xy'' + (2x + 1)y' + (x + 1)y = 0$
- (2) $xy'' + (1 - 2x)y' + (x - 1)y = 0$
- (3) $x^2(x^2 - 1)y'' - x(1 + x^2)y' + (1 + x^2)y = 0$
- (4) $x(x - 1)y'' + 2(2x - 1)y' + 2y = 0$

Solution of (1). Given

$$(0.16) \quad xy'' + (2x + 1)y' + (x + 1)y = 0$$

implies

$$y'' + \frac{(2x + 1)}{x}y' + \frac{(x + 1)}{x}y = 0$$

comparing with

$$y'' + \frac{b(x)}{x}y' + \frac{c(x)}{x^2}y = 0$$

we get

$$b(x) = 2x + 1 \text{ and } c(x) = x(x + 1)$$

and

$$b_0 = \lim_{x \rightarrow 0} b(x) = 1 \text{ and } c_0 = \lim_{x \rightarrow 0} c(x) = 0$$

Above step can be verified using following facts:

- As discussed in class b_0 and c_0 are coefficients of x^0 in power series expansion of $b(x)$ and $c(x)$ about origin, respectively.
- Since $b(x)$ and $c(x)$ are analytic at origin.

$$\Rightarrow b(x) = \sum_{n=0}^{\infty} b_n x^n = b_0 + b_1 x + b_2 x^2 + \cdots \text{ and } c(x) = \sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_3 x^2 + \cdots$$

$$\Rightarrow \lim_{x \rightarrow 0} b(x) = \lim_{x \rightarrow 0} (b_0 + b_1 x + b_2 x^2 + \cdots) = b_0 \text{ and } \lim_{x \rightarrow 0} c(x) = \lim_{x \rightarrow 0} (c_0 + c_1 x + c_3 x^2 + \cdots) = c_0$$

Thus the indicial equation is given by

$$\begin{aligned} r(r-1) + b_0 r + c_0 &= 0 \\ \Rightarrow r(r-1) + r &= 0 \\ \Rightarrow r^2 &= 0 \\ \Rightarrow r &= 0, 0 \end{aligned}$$

Thus solutions y_1 and y_2 are given by

$$y_1 = \sum_{m=0}^{\infty} a_m x^m$$

$$y_2 = y_1(x) \log x + \sum_{m=1}^{\infty} A_m x^m, \quad x > 0.$$

Putting $y = \sum_{m=0}^{\infty} a_m x^m$, $y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$ and $y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$ in equation (0.16) we get:

$$\sum_{m=2}^{\infty} m(m-1) a_m x^{m-1} + 2 \sum_{m=1}^{\infty} m a_m x^m + \sum_{m=1}^{\infty} m a_m x^{m-1} + \sum_{m=0}^{\infty} a_m x^{m+1} + \sum_{m=0}^{\infty} a_m x^m = 0$$

Substituting $m \rightarrow m+1$ in the first series, $m \rightarrow m+1$ in the third series and $m \rightarrow m-1$ in the fourth series, we get

$$\begin{aligned} \Rightarrow \sum_{m=1}^{\infty} m(m+1) a_{m+1} x^m + 2 \sum_{m=1}^{\infty} m a_m x^m + \sum_{m=0}^{\infty} (m+1) a_{m+1} x^m + \sum_{m=1}^{\infty} a_{m-1} x^m + \sum_{m=0}^{\infty} a_m x^m &= 0 \\ \Rightarrow \sum_{m=1}^{\infty} \{(m+1)^2 a_{m+1} + (2m+1) a_m + a_{m-1}\} x^m + a_0 + a_1 &= 0 \end{aligned}$$

Comparing coefficients of x^0 , we get

$$a_1 = -a_0,$$

Comparing coefficients of x^m , we get

$$a_{m+1} = -\frac{(2m+1)a_m + a_{m-1}}{(m+1)^2}, m \geq 1$$

For $m=1$,

$$a_2 = -\frac{3a_1 + a_0}{2^2} = \frac{2a_0}{2^2} = \frac{a_0}{2!}$$

For $m=2$,

$$a_3 = -\frac{5a_2 + a_1}{3^2} = -\frac{10a_0 - 4a_0}{2^2 3^2} = -\frac{6a_0}{2^2 \cdot 3^2} = -\frac{a_0}{3!}$$

For $m=3$,

$$a_4 = -\frac{7a_3 + a_2}{4^2} = -\frac{-\frac{7a_0}{3!} + \frac{a_0}{2!}}{4^2} = \frac{7a_0 - 3a_0}{4 \cdot 4!} = \frac{a_0}{4!}$$

for general $m \geq 1$, we have

$$a_m = \frac{(-1)^m}{m!} a_0$$

Therefore a solution is given by

$$y = \sum_{m=0}^{\infty} \frac{(-1)^m}{m!} a_0 x^m = a_0 e^{-x}.$$

To get second linearly independent soln. we use **the method of reduction of order**. Thus

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} e^{-\int p \, dx} dx \quad \text{where } p = \frac{2x+1}{x} \\ &= e^{-x} \int \frac{1}{e^{-2x}} e^{-\int \frac{(2x+1)}{x} dx} dx \\ &= e^{-x} \int e^{2x} \cdot e^{-2x} \cdot \frac{1}{x} dx \\ &= e^{-x} \log x \end{aligned}$$

Hence a basis of solutions of

$$xy'' + (2x+1)y' + (x+1)y = 0$$

is

$$y_1 = e^{-x} \text{ and } y_2 = e^{-x} \log x.$$

Solution of (2). Given

$$(0.17) \quad xy'' + (1-2x)y' + (x-1)y = 0$$

implies

$$y'' + \frac{(1-2x)}{x}y' + \frac{(x-1)}{x}y = 0$$

comparing with

$$y'' + \frac{b(x)}{x}y' + \frac{c(x)}{x^2}y = 0$$

we get

$$b(x) = (1-2x) \text{ and } c(x) = x(x-1)$$

and

$$b_0 = \lim_{x \rightarrow 0} b(x) = 1 \text{ and } c_0 = \lim_{x \rightarrow 0} c(x) = 0$$

Thus the indicial equation is given by

$$\begin{aligned} r(r-1) + b_0r + c_0 &= 0 \\ \Rightarrow r(r-1) + r &= 0 \\ \Rightarrow r^2 &= 0 \\ \Rightarrow r &= 0, 0 \end{aligned}$$

Thus y_1 and y_2 are given by

$$y_1 = \sum_{m=0}^{\infty} a_m x^m$$

$$y_2 = y_1(x) \log x + \sum_{m=1}^{\infty} A_m x^m, \quad x > 0.$$

Putting $y = \sum_{m=0}^{\infty} a_m x^m$, $y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$ and $y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$ in equation (0.17) we get:

$$\sum_{m=2}^{\infty} m(m-1) a_m x^{m-1} - 2 \sum_{m=1}^{\infty} m a_m x^m + \sum_{m=1}^{\infty} m a_m x^{m-1} + \sum_{m=0}^{\infty} a_m x^{m+1} - \sum_{m=0}^{\infty} a_m x^m = 0$$

Substituting $m \rightarrow m+1$ in the first series, $m \rightarrow m+1$ in the third series and $m \rightarrow m-1$ in the fourth series, we get

$$\begin{aligned} \Rightarrow \sum_{m=1}^{\infty} m(m+1) a_{m+1} x^m - 2 \sum_{m=1}^{\infty} m a_m x^m + \sum_{m=0}^{\infty} (m+1) a_{m+1} x^m + \sum_{m=1}^{\infty} a_{m-1} x^m - \sum_{m=0}^{\infty} a_m x^m &= 0 \\ \Rightarrow \sum_{m=1}^{\infty} \{(m+1)^2 a_{m+1} - (2m+1) a_m + a_{m-1}\} x^m - a_0 + a_1 &= 0. \end{aligned}$$

Comparing coefficients of x^0 , we get

$$a_1 = a_0,$$

Comparing coefficients of x^m , we get

$$a_{m+1} = \frac{(2m+1)a_m - a_{m-1}}{(m+1)^2}, m \geq 1$$

For $m = 1$,

$$a_2 = \frac{3a_1 - a_0}{2^2} = \frac{2a_0}{2^2} = \frac{a_0}{2!}$$

For $m = 2$,

$$a_3 = \frac{5a_2 - a_1}{3^2} = \frac{10a_0 - 4a_0}{2^2 \cdot 3^2} = \frac{6a_0}{2^2 \cdot 3^2} = \frac{a_0}{3!}$$

For $m = 3$,

$$a_4 = \frac{7a_3 - a_2}{4^2} = \frac{\frac{7a_0}{3!} - \frac{a_0}{2!}}{4^2} = \frac{7a_0 - 3a_0}{4 \cdot 4!} = \frac{a_0}{4!}$$

for general $m \geq 1$, we have

$$a_m = \frac{1}{m!} a_0$$

Therefore a solution is given by

$$y = \sum_{m=0}^{\infty} \frac{1}{m!} a_0 x^m = a_0 e^x.$$

To get second linearly independent solution we use **the method of reduction of order**. Thus

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} e^{-\int p dx} dx \quad \text{where } p = \frac{1-2x}{x} \\ &= e^x \int \frac{1}{e^{2x}} e^{-\int \frac{(1-2x)}{x} dx} dx \\ &= e^x \int e^{-2x} \cdot e^{2x} \cdot \frac{1}{x} dx \\ &= e^x \log x \end{aligned}$$

Hence a basis of solutions of

$$xy'' + (1-2x)y' + (x-1)y = 0$$

is

$$y_1 = e^x \text{ and } y_2 = e^x \log x.$$

Solution of (3). Given

$$(0.18) \quad x^2(x^2-1)y'' - x(1+x^2)y' + (1+x^2)y = 0$$

comparing with

$$y'' + \frac{b(x)}{x} y' + \frac{c(x)}{x^2} y = 0$$

we get

$$b(x) = -\frac{1+x^2}{x^2-1} \text{ and } c(x) = \frac{1+x^2}{x^2-1}$$

and

$$b_0 = \lim_{x \rightarrow 0} b(x) = 1 \text{ and } c_0 = \lim_{x \rightarrow 0} c(x) = -1$$

Thus the indicial equation is given by

$$\begin{aligned} r(r-1) + b_0r + c_0 &= 0 \\ \Rightarrow r(r-1) + r - 1 &= 0 \\ \Rightarrow r^2 &= 1 \\ \Rightarrow r &= 1, -1 \end{aligned}$$

Thus solutions y_1 and y_2 are given by

$$y_1 = x \sum_{m=0}^{\infty} a_m x^m$$

$$y_2 = k y_1(x) \log x + \sum_{m=1}^{\infty} A_m x^{m-1}, \quad x > 0.$$

Putting $y = \sum_{m=0}^{\infty} a_m x^{m+1}$, $y' = \sum_{m=0}^{\infty} (m+1)a_m x^m$ and $y'' = \sum_{m=1}^{\infty} (m+1)ma_m x^{m-1}$ in equation (0.18) we get:

$$\begin{aligned} x^2(x^2-1) \sum_{m=1}^{\infty} (m+1)ma_m x^{m-1} - x(1+x^2) \sum_{m=0}^{\infty} (m+1)a_m x^m + (1+x^2) \sum_{m=0}^{\infty} a_m x^{m+1} &= 0 \\ \Rightarrow \sum_{m=1}^{\infty} (m+1)ma_m x^{m+3} - \sum_{m=1}^{\infty} (m+1)ma_m x^{m+1} - \sum_{m=0}^{\infty} (m+1)a_m x^{m+1} - \sum_{m=0}^{\infty} (m+1)a_m x^{m+3} \\ + \sum_{m=0}^{\infty} a_m x^{m+1} + \sum_{m=0}^{\infty} a_m x^{m+3} &= 0 \end{aligned}$$

Substituting $m \rightarrow m-2$ in the first, fourth and sixth series, we get

$$\begin{aligned} \Rightarrow \sum_{m=3}^{\infty} (m-1)(m-2)a_{m-2}x^{m+1} - \sum_{m=1}^{\infty} (m+1)ma_m x^{m+1} - \sum_{m=0}^{\infty} (m+1)a_m x^{m+1} - \sum_{m=2}^{\infty} (m-1)a_{m-2}x^{m+1} \\ + \sum_{m=0}^{\infty} a_m x^{m+1} + \sum_{m=2}^{\infty} a_{m-2}x^{m+1} &= 0 \\ \Rightarrow \sum_{m=3}^{\infty} \{(m-1)(m-2)a_{m-2} - (m+1)ma_m - (m+1)a_m - (m-1)a_{m-2} + a_m + a_{m-2}\}x^{m+1} \\ + \{-2a_1x^2 - 6a_2x^3\} + \{-a_0x - 2a_1x^2 - 3a_2x^3\} + \{-a_0x^3\} + \{a_0x + a_1x^2a_2x^3\} + \{a_0x^3\} &= 0 \end{aligned}$$

Comparing co-efficients of x^3 in above equation, we get

$$a_2 = 0$$

Comparing co-efficients of x^2 in above equation, we get

$$a_1 = 0$$

Comparing co-efficients of x^1 in above equation, we get

$$a_0 \text{ is arbitrary.}$$

Comparing co-efficients of $x^m, m \geq 3$ in above equation, we get

$$a_m = \frac{(m-2)^2}{m(m+2)}a_{m-2}$$

Letting $m = 3, 5, \dots, 2k - 1$ in above recurrence relation, we can observe that

$$a_m = K_m a_1 = 0$$

In the same way, letting $m = 4, 6, 8, \dots, 2k$ in above recurrence relation, we obtain

$$a_m = P_m a_2 = 0$$

Hence

$$a_m = 0, \forall m \geq 1$$

Hence solution of (0.18) is given by

$$y_1(x) = a_0 x, \text{ where } a_0 \text{ is arbitrary constant.}$$

To get second linearly independent soln. we use **the method of reduction of order**. Thus

$$\begin{aligned} y_2 &= y_1 \int \frac{1}{y_1^2} e^{-\int p dx} dx \text{ where } p = -\frac{(1+x^2)}{x(x^2-1)} = \frac{1}{x} - \frac{2x}{x^2-1} \text{ and } \int p(x) dx = \ln|x| - \ln|x^2-1| = \ln\left|\frac{x}{x^2-1}\right| \\ &= x \int \frac{1}{x^2} \frac{x^2-1}{x} \\ &= x \int \frac{x^2-1}{x^3} dx \\ &= x \int \frac{1}{x} dx - x \int \frac{1}{x^3} \\ &= x \ln x + \frac{1}{2x} \end{aligned}$$

Hence a basis of solution of

$$x^2(x^2-1)y'' - x(1+x^2)y' + (1+x^2)y = 0$$

is

$$y_1 = x \text{ and } y_2 = x \ln x + \frac{1}{2x}.$$

Solution of (4).

(0.19)

$$x(x-1)y'' + 2(2x-1)y' + 2y = 0$$

comparing with

$$y'' + \frac{b(x)}{x}y' + \frac{c(x)}{x^2}y = 0$$

we get

$$b(x) = \frac{2(2x-1)}{x-1} \text{ and } c(x) = \frac{2x}{x-1}$$

and

$$b_0 = \lim_{x \rightarrow 0} b(x) = 2 \text{ and } c_0 = \lim_{x \rightarrow 0} c(x) = 0$$

Thus the indicial equation is given by

$$\begin{aligned} r(r-1) + b_0 r + c_0 &= 0 \\ \Rightarrow r(r-1) + 2r &= 0 \\ \Rightarrow r^2 + r &= 0 \\ \Rightarrow r &= 0, -1 \end{aligned}$$

Thus solutions y_1 and y_2 are given by

$$\begin{aligned} y_1 &= \sum_{m=0}^{\infty} a_m x^m \\ y_2 &= k y_1(x) \log x + \sum_{m=1}^{\infty} A_m x^{m-1}, \quad x > 0. \end{aligned}$$

Putting $y = \sum_{m=0}^{\infty} a_m x^m$, $y' = \sum_{m=1}^{\infty} m a_m x^{m-1}$ and $y'' = \sum_{m=2}^{\infty} m(m-1) a_m x^{m-2}$ in equation (0.19) we get:

$$\begin{aligned}
& x(x-1) \sum_{m=2}^{\infty} m(m-1)a_m x^{m-2} + 2(2x-1) \sum_{m=1}^{\infty} ma_m x^{m-1} + 2 \sum_{m=0}^{\infty} a_m x^m = 0 \\
\Rightarrow & \sum_{m=2}^{\infty} m(m-1)a_m x^m - \sum_{m=2}^{\infty} m(m-1)a_m x^{m-1} + 4 \sum_{m=1}^{\infty} ma_m x^m - 2 \sum_{m=1}^{\infty} ma_m x^{m-1} + 2 \sum_{m=0}^{\infty} a_m x^m = 0 \\
& \text{Substituting } m \rightarrow m+1 \text{ in the second and fourth series, we get} \\
\Rightarrow & \sum_{m=2}^{\infty} m(m-1)a_m x^m - \sum_{m=1}^{\infty} (m+1)ma_{m+1}x^m + 4 \sum_{m=1}^{\infty} ma_m x^m - 2 \sum_{m=0}^{\infty} (m+1)a_{m+1}x^m + 2 \sum_{m=0}^{\infty} a_m x^m = 0 \\
\Rightarrow & \sum_{m=2}^{\infty} \{m(m-1)a_m - (m+1)ma_{m+1} + 4ma_m - 2(m+1)a_{m+1} + 2a_m\}x^m \\
& + \{-2a_2x\} + \{4a_1x\} + \{-2a_1 - 4a_2x\} + \{2a_0 + 2a_1x\} = 0
\end{aligned}$$

Comparing co-efficients of x^0 in above equation , we get

$$a_1 = a_0,$$

Comparing co-efficients of x^1 in above equation , we get

$$a_2 = a_1 = a_0,$$

Comparing co-efficients of $x^m, m \geq 2$ in above equation , we get

$$a_{m+1} = \frac{(m+2)(m+1)}{(m+2)(m+1)}a_m \Rightarrow a_{m+1} = a_m = a_0$$

Hence solution of (0.19) is given by

$$y_1(x) = a_0(1 + x + x^2 + x^3 + \cdots) = a_0 \frac{1}{1-x}$$

To get second linearly independent soln. we use **the method of reduction of order**. Thus

$$\begin{aligned}
y_2 &= y_1 \int \frac{1}{y_1^2} e^{-\int p dx} dx \quad \text{where } p = \frac{2(2x-1)}{x(x-1)} \quad \text{and } \int p(x) dx = 2 \ln |x^2 - x| \\
&= \frac{1}{1-x} \int (1-x)^2 \frac{1}{(x^2-x)^2} dx \\
&= \frac{1}{1-x} \int \frac{1}{x^2} dx \\
&= -\frac{1}{1-x} \frac{1}{x} \\
&= -\frac{1}{x(1-x)}
\end{aligned}$$

Hence a basis of solutions of

$$x(x-1)y'' + 2(2x-1)y' + 2y = 0$$

is

$$y_1 = \frac{1}{1-x} \quad \text{and} \quad y_2 = \frac{1}{x(1-x)}.$$

Example 0.8. Show that between any two consecutive positive zeros of $J_n(x)$ there is precisely one zero of $J_{n+1}(x)$ and vice versa.

Solution. Let $0 < x_1 < x_2$ be two consecutive positive zeros of $J_n(x)$, that is,

$$J_n(x_1) = 0, J_n(x_2) = 0 \text{ and } J_n(x) \neq 0 \forall x \in (x_1, x_2).$$

$\Rightarrow x^n J_n(x)$ and $x^{-n} J_n(x)$ are zero for $x = x_1$ & $x = x_2$.

$\Rightarrow$ By Rolle's Theorem $\exists x_3, x_4 \in (x_1, x_2)$ such that

$$[x_3^n J_n(x_3)]' = x_3^n J_{n-1}(x_3) = 0$$

and

$$[x_4^{-n} J_n(x_4)]' = -x_4^{-n} J_{n+1}(x_4) = 0$$

$\Rightarrow$ Between two zeros of J_n , we have at least one zero of both J_{n-1} & J_{n+1} .

To prove uniqueness, let there exist two zero of J_{n+1} between consecutive zeros x_1 and x_2 of J_n . This implies that there exist a zero of J_n between x_1 and x_2 , which contradicts the fact that x_1 and x_2 are consecutive zeroes.

Example 0.9. Find a general solution in terms of J_ν and $J_{-\nu}$ or indicate when this is not possible. Use the indicated substitutions.

(1) $x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0$, with $\lambda x = z$

(2) $x^2 y'' + xy' + \frac{1}{4}(x^2 - 1)y = 0$, with $x = 2z$

(3) $xy'' + (2\nu + 1)y' + xy = 0$, with $y = x^{-\nu}u$

Solution of (1). Given

$$(0.20) \quad x^2 y'' + xy' + (\lambda^2 x^2 - \nu^2)y = 0$$

Put

$$z = \lambda x \Rightarrow \frac{dz}{dx} = \lambda$$

By Chain Rule, we obtain

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \lambda \frac{dy}{dz}$$

Similarly,

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\lambda \frac{dy}{dz} \right) = \lambda \frac{d^2 y}{dz^2} \frac{dz}{dx} = \lambda^2 \frac{d^2 y}{dz^2}$$

We may rewrite equation (0.20) as the following:

$$(0.21) \quad \begin{aligned} & \lambda^2 x^2 \frac{1}{\lambda^2} \frac{d^2 y}{dz^2} + \lambda x \frac{1}{\lambda} \frac{dy}{dz} + (\lambda^2 x^2 - \nu^2)y = 0 \\ & \Rightarrow z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + (z^2 - \nu^2)y = 0 \end{aligned}$$

This is Bessel's equation of order ν .

The solution of (0.21) is given by

$$y(z) = C_1 J_\nu(z) + C_2 J_{-\nu}(z)$$

implies

$$y(x) = C_1 J_\nu(\lambda x) + C_2 J_{-\nu}(\lambda x) \text{ is a solution of (0.20).}$$

Solution of (2). Given

$$(0.22) \quad x^2 y'' + xy' + \frac{1}{4}(x^2 - 1)y = 0$$

Substitute

$$x = 2z \Rightarrow \frac{dz}{dx} = \frac{1}{2}.$$

By Chain Rule, we obtain

$$\frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \frac{1}{2} \frac{dy}{dz}$$

Similarly,

$$\frac{d^2 y}{dx^2} = \frac{d}{dx} \left(\frac{1}{2} \frac{dy}{dz} \right) = \frac{1}{2} \frac{d^2 y}{dz^2} \frac{dz}{dx} = \frac{1}{4} \frac{d^2 y}{dz^2}$$

We may rewrite equation (0.22) as the following:

$$\begin{aligned} \frac{x^2}{4} \cdot 4 \frac{d^2 y}{dx^2} + \frac{x}{2} 2 \frac{dy}{dx} + \left(\frac{x^2}{4} - \frac{1}{4} \right) y &= 0 \\ (0.23) \quad \Rightarrow z^2 \frac{d^2 y}{dz^2} + z \frac{dy}{dz} + \left(z^2 - \left(\frac{1}{2} \right)^2 \right) y &= 0 \end{aligned}$$

This is Bessel equation of order $\frac{1}{2}$.

Thus the solution of (0.23) is given by

$$y = c_1 J_{\frac{1}{2}}(z) + c_2 J_{-\frac{1}{2}}(z)$$

Hence

$$y = c_1 J_{\frac{1}{2}}\left(\frac{x}{2}\right) + c_2 J_{-\frac{1}{2}}\left(\frac{x}{2}\right).$$

is a solution of (0.22)

Solution of (3). Given

$$(0.24) \quad xy'' + (2\nu + 1)y' + xy = 0$$

$$\begin{aligned} u &= x^\nu y \\ \Rightarrow u' &= \nu x^{\nu-1} y + x^\nu y' \\ \Rightarrow u'' &= \nu(\nu-1)x^{\nu-2}y + 2\nu x^{\nu-1}y' + x^\nu y'' \end{aligned}$$

Substitute $y'' = -x^{-1}(2\nu+1)y' - y$ in the last term of above u'' - equation, we get

$$\begin{aligned} u'' &= \nu(\nu-1)x^{\nu-2}y + 2\nu x^{\nu-1}y' + x^\nu \{-x^{-1}(2\nu+1)y' - y\} \\ &= \nu(\nu-1)x^{\nu-2}y + 2\nu x^{\nu-1}y' - 2\nu x^{\nu-1}y' - x^{\nu-1}y' - x^\nu y \\ &= \nu(\nu-1)x^{\nu-2}y - x^{\nu-1}y' - x^\nu y \end{aligned}$$

Again, substitute $x^{\nu-1}y' = x^{-1}u' - \nu x^{\nu-2}y$ in the second term of above u'' - equation, we get

$$\begin{aligned} u'' &= \nu(\nu-1)x^{\nu-2}y - \{x^{-1}u' - \nu x^{\nu-2}y\} - x^\nu y \\ &= \nu^2 x^{\nu-2}y - \nu x^{\nu-2}y - \frac{1}{x}u' + \nu x^{\nu-2}y - x^\nu y \\ &= -\frac{1}{x}u' - x^{\nu-2}y(x^2 - \nu^2) \end{aligned}$$

Again, substitute $x^{\nu-2}y = x^{-2}u$ in the second term of above u'' - equation, we get

$$\begin{aligned} u'' &= -\frac{1}{x}u' - \frac{1}{x^2}(x^2 - \nu^2)u \\ (0.25) \quad \Rightarrow x^2 u'' + x u' + (x^2 - \nu^2)u &= 0 \end{aligned}$$

This is Bessel's equation of order $\nu \notin \mathbb{Z}$.

The solution of (0.25) is given by

$$u = C_1 J_\nu(x) + C_2 J_{-\nu}(x)$$

Therefore,

$$y = x^{-\nu} \{C_1 J_\nu(x) + C_2 J_{-\nu}(x)\}$$

is a solution of (0.24) .

Example 0.10. Using recurrence relations, show that

$$xJ'_{n+1}(x) + (n+1)J_{n+1}(x) = xJ_n(x).$$

Solution. Recall,

$$(0.26) \quad J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_{\nu}(x)$$

$$(0.27) \quad J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_{\nu}(x)$$

Adding (0.26)&(0.27) we get

$$(0.28) \quad J_{\nu-1}(x) = \frac{\nu}{x} J_{\nu}(x) + J'_{\nu}(x)$$

Put $\nu = n+1$ in equation (0.28) to get

$$\begin{aligned} J_n(x) &= \frac{n+1}{x} J_{n+1}(x) + J'_{n+1}(x) \\ \Rightarrow xJ'_{n+1}(x) + (n+1)J_{n+1}(x) &= xJ_n(x). \end{aligned}$$

Example 0.11. Show that the Bessel functions J_{ν} ($\nu \geq 0$) satisfy

$$\int_0^1 xJ_{\nu}(\lambda_m x)J_{\nu}(\lambda_n x) dx = \frac{1}{2}J_{\nu+1}^2(\lambda_n) \delta_{mn},$$

where λ_n are the positive zeros of J_{ν} .

Solution. We know that $y(t) = J_{\nu}(t)$ satisfies

$$\ddot{y} + \frac{1}{t}\dot{y} + (1 - \frac{\nu^2}{t^2})y = 0, \quad \equiv \frac{d}{dt}$$

Let $t = \lambda x \Rightarrow y(t) = y(\lambda x) = u(x)$. Then $u'(x) = \lambda \dot{y}$ and $u''(x) = \lambda^2 \ddot{y}$. Hence $u(x) = J_{\nu}(\lambda x)$ satisfies

$$(0.29) \quad u'' + \frac{1}{x}u' + (\lambda^2 - \frac{\nu^2}{x^2})u = 0,$$

Similarly, $v(x) = J_{\nu}(\mu x)$ satisfies

$$(0.30) \quad v'' + \frac{1}{x}v' + (\mu^2 - \frac{\nu^2}{x^2})v = 0.$$

Multiplying (0.29) by v and (0.30) by u and subtracting, we find

$$\frac{d}{dx}[x(u'v - uv')] = (\mu^2 - \lambda^2)xuv.$$

Integrating from $x = 0$ to $x = 1$ we find

$$(0.31) \quad (\mu^2 - \lambda^2) \int_0^1 xuv dx = u'(1)v(1) - u(1)v'(1).$$

Now $u(1) = J_{\nu}(\lambda)$ and $v(1) = J_{\nu}(\mu)$. Let us choose $\lambda = \lambda_m$ and $\mu = \lambda_n$, where λ_m and λ_n are positive zeros of J_{ν} . Then $u(1) = v(1) = 0$ and thus find

$$(\lambda_n^2 - \lambda_m^2) \int_0^1 xJ_{\nu}(\lambda_m x)J_{\nu}(\lambda_n x)dx = 0.$$

If $n \neq m$ then

$$\int_0^1 xJ_{\nu}(\lambda_m x)J_{\nu}(\lambda_n x)dx = 0.$$

Now from (0.31), we find [since $u'(x) = \lambda J'_{\nu}(\lambda x)$ etc]

$$\int_0^1 xJ_{\nu}^2(\lambda x)dx = \lim_{\mu \rightarrow \lambda} \frac{\lambda J'_{\nu}(\lambda)J_{\nu}(\mu) - \mu J_{\nu}(\lambda)J'_{\nu}(\mu)}{\mu^2 - \lambda^2}$$

$$= \frac{\lambda(J'_\nu(\lambda))^2 - J_\nu(\lambda)J'_\nu(\lambda) - \lambda J_\nu(\lambda)J''_\nu(\lambda)}{2\lambda}$$

Now if we take $\lambda = \lambda_n$ where λ_n is a positive zero of J_ν , then we find

$$\int_0^1 x J_\nu^2(\lambda_n x) dx = \frac{1}{2}(J'_\nu(\lambda_n))^2.$$

Now from

$$(x^{-\nu} J_\nu(x))' = -x^{-\nu} J_{\nu+1}(x) \Rightarrow J'_\nu(x) - \frac{\nu}{x} J_\nu(x) = -J_{\nu+1}(x),$$

we find by substituting $x = \lambda_n$

$$J'_\nu(\lambda_n) = -J_{\nu+1}(\lambda_n).$$

Thus, finally we get

$$\int_0^1 x J_\nu^2(\lambda_n x) dx = \frac{1}{2} J_{\nu+1}^2(\lambda_n).$$

Property 1. $[x^\nu J_\nu(x)]' = x^\nu J_{\nu-1}(x)$

Proof.

$$\begin{aligned} J_\nu(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+\nu}}{2^{2m+\nu} \cdot m! \cdot \Gamma(m+\nu+1)} \\ x^\nu J_\nu(x) &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m+2\nu}}{2^{2m+\nu} \cdot m! \cdot \Gamma(m+\nu+1)} \\ \text{Differentiating w.r.t } x \text{ we get} \\ [x^\nu J_\nu(x)]' &= \sum_{m=0}^{\infty} \frac{(-1)^m (2m+2\nu) x^{2m+2\nu-1}}{2 \cdot 2^{2m+\nu-1} \cdot m! \cdot (m+\nu) \Gamma(m+\nu)} \\ &= x^{2\nu-1} \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{2^{2m+\nu-1} \cdot m! \cdot \Gamma(m+\nu)} \\ &= x^\nu J_{\nu-1}(x) \end{aligned}$$

Similarly,

Property 2. $[x^{-\nu} J_\nu(x)]' = -x^{-\nu} J_{\nu+1}(x)$

Property 3. $J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$

Property 4. $J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x)$

Proof of property 3 and 4.

we have

$$(0.32) \quad \begin{aligned} [x^\nu J_\nu(x)]' &= x^\nu J_{\nu-1}(x) \\ \nu x^{\nu-1} J_\nu(x) + x^\nu J'_\nu(x) &= x^\nu J_{\nu-1}(x) \end{aligned}$$

$$(0.33) \quad \begin{aligned} [x^{-\nu} J_\nu(x)]' &= -x^{-\nu} J_{\nu+1}(x) \\ -\nu x^{-\nu-1} J_\nu(x) + x^{-\nu} J'_\nu(x) &= -x^{-\nu} J_{\nu+1}(x) \\ \text{Multiply by } x^{2\nu} \\ -\nu x^{\nu-1} J_\nu(x) + x^\nu J'_\nu(x) &= -x^\nu J_{\nu+1}(x) \end{aligned}$$

(0.32) + (0.33) implies

$$J_{\nu-1}(x) - J_{\nu+1}(x) = 2J'_\nu(x)$$

and (0.32) - (0.33) implies

$$J_{\nu-1}(x) + J_{\nu+1}(x) = \frac{2\nu}{x} J_\nu(x)$$